

Covid19 and corporate India

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August 31, 2020

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Abstract

We use earnings calls reports of Indian companies to study who was talking about Covid-19 in Jan-Feb 2020 when the case-load was still low in India but had begun to spread in other countries. Using natural language processing tools, we construct a new firm-specific measure of exposure to the pandemic. We find that firms that were more exposed to the pandemic early on in 2020 fared worse on stock market returns when the 24 March lockdown was announced. Firms with lower profits, higher FX earnings, higher share of inventories, higher operating expenses, and greater reliance on trade credit were also more affected around the first lockdown event. In ongoing work, we expand our sample to the universe of listed firms in India in order to explore the heterogeneities of responses to the national lockdown. We also analyse the impact on these firms of the unscheduled RBI announcement which took place three days after the lockdown announcement.

JEL Classification: G14; G18; G32; L25

Keywords: Covid-19; Corporate finance; Equity markets; Financial constraints; Public health; Non-financial firms

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This paper was written while Mathur was at IHEID, Geneva. We are grateful to comments from Shekhar Hari Kumar and Ajay Shah. The views expressed here are those of the authors and do not necessarily reflect those of the Bank of England or any of its committees. All errors are our own.

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1 Introduction

We are in the middle of a global pandemic, which is inflicting two kinds of shocks on countries: a health shock and an economic shock. The measures adopted to prevent the spread of the Coronavirus Disease 2019 (Covid-19), especially social distancing and lockdown, are inflicting severe damages on economies all around the world. One of the most stringent nationwide lockdowns was imposed in India on 24 March 2020. The pandemic along with the lockdown have dealt a massive shock to the Indian economy which has not spared almost any sector.

The 24 March lockdown in India was regarded as one of the most severe lockdowns in the world, based on data from the Oxford COVID-19 Government Response Tracker. All transport services, except those for essential personnel, were suspended, in addition to all educational, commercial, and private establishments. As a result of the lockdown, all non-essential expenditures got postponed causing aggregate demand to collapse across the country. There have been large scale disruptions to the supply chains as well.¹

The twin shocks of demand and supply disruption were a severe blow for the non-financial firms. As firms deal with labour disruptions, constrained availability of financing, and shrinking of markets for their goods and services, many among them continue to struggle to survive. They have to pay rents, salaries, debts, and maintain inventories even as their revenues have been steadily falling as people have changed lifestyles and preferences and cut back on expenditures.

The lockdown affected all sectors of the Indian economy. The stock market reacted negatively overall. This is not surprising, since stock prices reflect changes in expected future cash flows and/or discount rates. However it is possible that some firms were more affected than the others depending on their exposure to the pandemic as well as pre-pandemic characteristics.

In this paper, we propose a novel approach to identify firms that may have had greater exposure to the pandemic even before it assumed serious proportions in India, by virtue of, for example, their connections to other affected countries, among others. Specifically we propose the use of earnings call transcripts as an important source of information for gauging a firm’s fundamental exposure to the pandemic. We use natural language processing techniques and interpret the number of times a firm mentioned Covid-19 related words in its call reports as an indicator of its exposure to the pandemic.

Earnings calls typically follow the presentation of a firm’s quarterly results. These calls are attended by senior management of the firm (for example, the CEO, CFO, MD, etc), who present short prepared remarks, and then open the floor to questions from analysts. This implies that the calls are more spontaneous as compared to say the firm’s annual report, because the senior management answers questions on the fly from the audience. These reports therefore convey not just fundamental financial information, but also analysts’ and managers’ opinion about the firm (Borochin *et al.*, 2018).

A significant part of the literature focuses on the tone and sentiment of these reports and

¹Since the Indian government announced multiple lockdowns in the subsequent months, in our paper we refer to the March 24 lockdown interchangeably as the ‘first lockdown’.

their implications for stock market returns, trading volumes (Frankel *et al.*, 1999; Bushee *et al.*, 2003, 2004), and options pricing (Borochin *et al.*, 2018). Our work closely relates to two recent papers, Hassan *et al.* (2020) and Ramelli and Wagner (2020). Both these studies use the information contained in earnings call transcripts.

Hassan *et al.* (2020) focus on globally listed firms, and study whether firms that were more exposed to previous disease outbreaks such as SARS and MERS were better prepared for the 2020 pandemic, and therefore had higher equity returns, than those who were not. Ramelli and Wagner (2020) analyse characteristics of US firms that explain both their stock market performance between January and March 2020 and their discussions of Covid-19 in the earnings transcripts.

Unlike these papers, we use the information in earnings call reports, to measure fundamental exposure of Indian firms to the pandemic. We are interested in using this information to study the equity market performance of the firms around the largest, most stringent lockdown announced in the world (at the time).

We find that the occurrences of Covid-19 related words in the firms' earnings call reports were higher in those sectors which presumably have more fundamental exposure to the pandemic, for example in the form of supply-chains with China or other early-affected countries. Some of these sectors are pharmaceuticals, consumer goods, automobile, chemicals etc. On the other hand firms in health care services, financial services, media and entertainment, power and telecom industries either did not mention or mentioned much less pandemic related words during this period in their call reports.

In the Indian context our analysis complements earlier work by Sane and Sharma (2020) who calculated the liquidity cover of listed firms in the face of large revenue shocks during the pandemic and Bansal *et al.* (2020) who also examine variations in the market valuation of firms on account of firm-specific characteristics during the pandemic. In this work, we take a more holistic view of firm-level vulnerabilities, examining the fundamental exposure to the pandemic, as well as the role of financial flexibilities, including liquidity. We also focus on one specific event – the 24 March lockdown – in order to obtain greater precision.

We next examine whether the firms that were more exposed to the pandemic early on, fared worse when the nationwide lockdown was announced. We analyse the pre-pandemic balance sheet characteristics that may have worsened the impact of the lockdown on some firms compared to the others. In particular, we measure the differential, cross-sectional responses of firms to the 24 March lockdown announcement, using high-frequency stock market data, and annual pre-pandemic firm-level balance sheet variables and apply the standard event study methodology.

We find that firms that were more exposed to the pandemic early on in 2020 fared worse on stock market returns when the 24 March lockdown was announced. Firms with lower profits, higher FX earnings, higher share of inventories, higher operating expenses, and greater reliance on trade credit were also more affected around the first lockdown event.

In ongoing work, we expand our sample to the universe of all listed firms in India in order to further explore the firm-level heterogeneities of responses to the national lockdown. In addition we also plan to analyse the impact of the unscheduled monetary policy announcement made by the Reserve Bank of India, on March 27, i.e. 3 days after the

lockdown, on our sample of listed firms. The objective is to throw light on which firms may have benefitted from the policy package announced by the RBI.² Our paper unfolds as follows. In section 2 we discuss our findings from the analysis of the firms' call reports from the period Jan-Feb, 2020. In section 3 we conduct an event study analysis to find out what kind of firms were more affected when the nationwide lockdown was announced on March 24. In section 4 we discuss our ongoing research wherein we are extending the event-study analysis to a larger sample of listed firms and also investigating the potential effect of the RBI's announcement of March 27. Finally, we summarise our current take-aways in section 5.

2 Analysis of firms' earnings call reports

We focus on earnings calls conducted by firms in January and February 2020, when the Covid-19 case load was still low in India but the pandemic had begun spreading in other countries. These are calls discussing the income statements of October-December, 2019 (Q3 FY20) and January-March, 2020 (Q4 FY20) respectively, of the Indian financial year.

When India reported its first case of the Covid-19 pandemic on 30 January, 2020, close to 7,700 people had been infected all over the world, majority being in China. Other countries such as the US, Australia, Germany, Japan, South Korea, UAE and HongKong had started reporting Covid-19 cases. By end February, it had morphed into a full blown public health crisis. The total number of infections globally had risen to more than 83,000, with a death toll of more than 2,800. While the disease was spreading rapidly in countries such as Italy, South Korea, France, US and Iran, these were still early days of the pandemic in India which had less than 10 confirmed cases.

Focusing on the call reports of Jan-Feb 2020 enables us to analyse the firms' exposure to the pandemic at a time when the disease was still at a nascent stage in India unlike say March when the spread of the pandemic had begun affecting most firms. It also allows for easier identification of firm exposure because it is not muddled by domestic policy interventions. For example, there were only 13 Covid-19 related notifications issued by the Indian government in February, compared to 266 in March, as listed by PRS Legislative Research. Hence, March onwards, the stock market performance of all firms was likely to be affected by these interventions over and above firm-specific concerns around the disease itself.

We start with a sample of the largest listed firms on the NIFTY500 index of the National Stock Exchange (NSE) of India. Of the 500 firms in the index, we have access to the call reports of 196 firms in January-February 2020, and of 90 firms in April-May 2020.

2.1 Which firms had exposure to Covid-19 in Jan-Feb 2020?

We interpret the number of times a firm mentioned Covid-19 related words in its call reports as an indicator of its exposure to the pandemic. Accordingly, we count the number of times Covid-19 and related words (such as "coronavirus", "pandemic", "ncov",

²While this section is currently work in progress, we hope to finish this analysis and the full paper by the end of September, 2020

"sarscov", "epidemic" etc) are mentioned in the quarterly earnings call reports of the 196 firms in January-February 2020, and also of the 90 firms in April-May 2020 for the sake of comparison. We briefly summarise our findings below.

Table 1 Overall mentions of Covid19 in Q3 and Q4 FY20

	Quarter	Total mentions	No. of reports	Avg. words per report
1	Jan-Feb 2020	189	63 ($N = 196$)	3
2	Apr-May 2020	2,781	90 ($N = 90$)	30.900

Note: This table presents the total (column 2) and average (column 4) number of Covid19-related mentions in the sample of earnings call reports. There are 196 reports for Jan-Feb 2020 and 90 reports for Apr-May 2020, from the initial sample of Nifty 500.

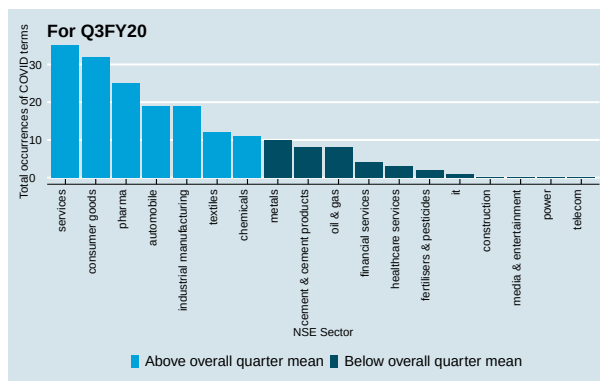
1. Only one-third of the firms in our Jan-Feb sample mention Covid-19 or related words. The average number of times these words are mentioned is three. Only three of the firms discussing the pandemic are in the financial services sector (table 1).
2. All 90 firms in the Apr-May 2020 sample mention Covid-19 or related words, demonstrating the extensive spread of the disease by this time. The average occurrence of the words per report is ten times higher, close to 31. This reflects our earlier concern that from March onwards, all firms had become exposed to the pandemic.
3. Even in Jan-Feb 2020, there were sector-wise heterogeneities in Covid-19 discussions, as shown in panel (a) of figure 1 below.
4. The occurrences of Covid-19 related words were higher in those sectors which presumably have more fundamental exposure to the pandemic, for example in the form of supply-chains with China or other early-affected countries. Some of these sectors are pharmaceuticals, consumer goods, automobile, chemicals etc.
5. Firms in health care services, financial services, media and entertainment, power and telecom industries either did not mention or mentioned much less pandemic related words during this period in their call reports. With the possible exception of health care, these sectors were likely to be affected due to indirect exposure to the pandemic.

We also look at the firms that mentioned “supply”, “demand” and “uncertainty” related words in context of the Covid-19 discussion in their call reports. These are likely to be the most common channels of disruption faced by the firms during the pandemic. In panels (b)-(d) of figure 1, we find that firms in sectors with higher than average mentions of Covid-19 related words also had higher than average mentions of “supply” related words in the sentences where Covid-19 was discussed. For firms in the services sector, mentions of "demand" related words in the context of the disease were higher. For all the sectors having higher than average mentions of Covid-19 related words, we find significantly higher mentions of "uncertainty" related words in the context of the pandemic.

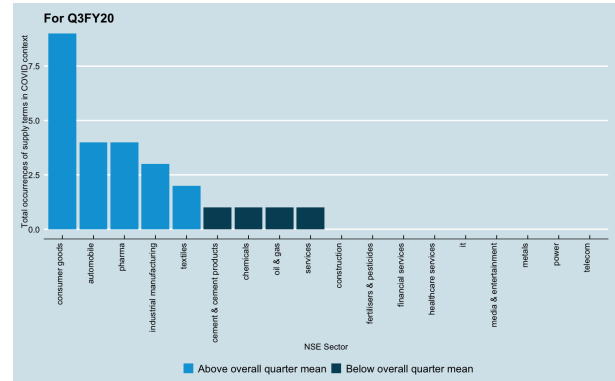
This preliminary analysis gives us an idea of which firms and sectors had greater exposure to the pandemic as early as Jan-Feb 2020 when the disease still hadn’t spread in India.

For subsequent analysis, we consider the firms that mention Covid-19 in Jan-Feb 2020

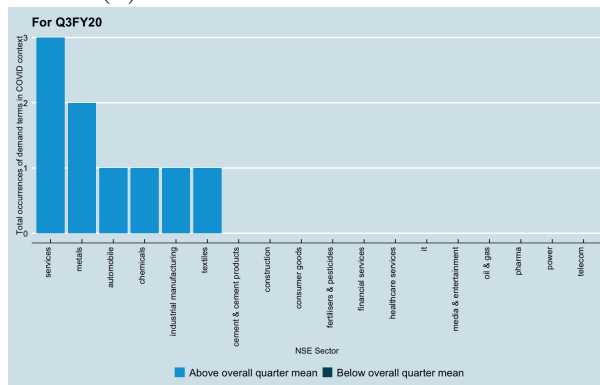
Figure 1 Sector-wise occurrences of Covid19 terms for Jan-Feb 2020



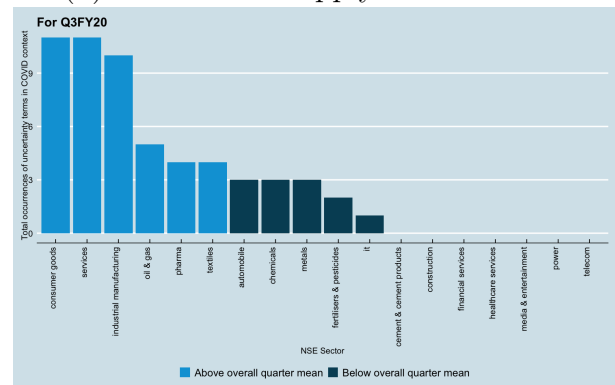
(a) Mentions of Covid19 terms



(b) Mentions of supply-related terms



(c) Mentions of demand-related terms



(d) Mentions of uncertainty-related terms

Note: All panels use NSE sector classification. Panel (a) shows the distribution of Covid-19 words. For the other three panels, we first extract sentences that are related to Covid-19, using the same keywords as those used in panel (a). Then, we construct frequencies of demand, supply, and uncertainty related words within those sentences. Panel (b) shows the distribution of supply-related words (such as “supply”, “supplies”, “supply-chains”, etc) in Covid-19 related sentences. Similarly, panels (c) and (d) show distribution of demand and uncertainty related terms in Covid-19 sentences.

as our “treated” sample and those that did not discuss the pandemic as our “control” sample. A relevant question to ask is how similar are the “treated” and “control” samples in terms of their key balance sheet characteristics. Using annual data from the pre-pandemic period (ending in March 2019), we compare the two sets of firms in size, age, profit, foreign exchange earnings, inventories, cash balances etc.

For ease of comparison, we drop the three firms that are in the financial services sector and that mentioned pandemic related words in the Jan-Feb call reports. As shown in table 2 below, we do not find any major difference in the characteristics of these two groups of firms, except that the “treated” firms are on average older and hold higher inventories than the “control” group firms.³

³All firm specific data has been collected from the Prowess database of the Centre for Monitoring Indian Economy (CMIE).

Table 2 Summary statistics for non-financial firms: Data as of March 31, 2019

Variable	Mean of firms with no COVID mentions “Control” N = 96	Mean of firms with COVID mentions “Treated” N = 60	p-value
Covid word-count	0	3.083	0
Scaled COVID word-count	0	0.023	0
Age	35.011	43.733	0.014
Size	11.187	11.125	0.787
Leverage	0.149	0.158	0.734
Profit	0.256	0.234	0.458
Trade ratio	0.490	0.066	0.483
FX ratio	0.300	0.265	0.514
Capital ratio	0.573	0.600	0.378
Cash ratio	0.071	0.049	0.117
Trade receivables ratio	0.139	0.129	0.538
Inventories ratio	0.089	0.124	0.060
Short-term borrowings ratio	0.491	0.432	0.369
Tangibles ratio	0.370	0.418	0.307
Operating expenses ratio	0.758	0.780	0.350

Note: This table presents the t-tests between firms that mention Covid19 in Jan-Feb 2020 (“treated”), and those that do not (“control”). We only consider non-financial firms here. The sample consists of the firms for whom we have earnings call transcripts for Jan-Feb 2020. The control firms are in column 2 (N = 96); while the treated firms are in column 3 (N = 60). It shows that in terms of the balance sheet metrics, the two sets of firms are quite similar except for their age and inventories.

2.2 What kind of exposure did firms have to the pandemic in Jan-Feb, 2020?

We next analyse the context within which the firms discussed the pandemic in their call reports, for example, references to supply-chains, demand disruptions, or uncertainty due to the pandemic and so on. This will give us a sense of the kind of exposure the firms may have had to the pandemic in the early part of 2020.

We use the techniques applied in [Mathur and Sengupta \(2019\)](#). For every firm’s call report, we first isolate the sentences that contain Covid-19 related words. There are 176 sentences in total for the Jan-Feb 2020 reports. Then, we create a word cloud with the most frequently occurring words in these sentences, after stripping out stop-words and other uninformative words.

The word cloud for the Jan-Feb 2020 reports is shown in figure 2. The size of each word is directly proportional to its frequency in the sentences. We plot the 50 most frequently occurring words. All the coronavirus related words, which are the most common words in these sentences by construction, are not plotted here for ease of comprehension.

1. The words “china” and “impact” occur most frequently indicating that firms were talking about the origin of the coronavirus disease and its effect.
2. We also see words related to areas where the impact of the pandemic was potentially anticipated or the expected transmission channels of the disease such as “earnings”, “shipping”, “pharma”, “macro”, “supply chain”, “trade”, “logistics”, “imports”, “demand”, “supply”, “prices” etc.

Figure 2 Word clouds of sentences with Covid-19 related words in Jan-Feb call reports



Note: The graph shows the fifty most frequent words in Covid19 related sentences. Uninformative words like stopwords etc. are purged from the sentences. “Coronavirus” and related-words have been removed before plotting for ease of comprehension.

3 Which firms were more affected by the 24 March lockdown announcement?

The 24 March lockdown in India was regarded as one of the most severe lockdowns in the world, based on data from the Oxford COVID-19 Government Response Tracker (Hale *et al.*, 2020). All transport services, except those for essential personnel, were suspended, in addition to all educational, commercial, and private establishments.⁴ The lockdown affected all sectors of the Indian economy. The stock market reacted negatively overall. This is not surprising, since stock prices reflect changes in expected future cash flows and/or discount rates. However it is possible that some firms were more affected than the others depending on their exposure to the pandemic as well as pre-pandemic characteristics.

To measure the differential, cross-sectional responses of firms to the lockdown announcement, we use high-frequency stock market data and an event study methodology. We have two main hypotheses.

Our **primary hypothesis** is that firms that were more exposed to the pandemic and mentioned Covid-19 in their earnings call reports in Jan-Feb 2020 (the "treated" group) fared worse than the "control" group when the lockdown was announced.

If investors believe that firms who discussed Covid-19 and its implications for their businesses early on in the year are more exposed to the virus, for example due to supply

⁴See: https://www.mha.gov.in/sites/default/files/Guidelines_0.pdf.

chains with China, or factories in badly-affected countries like Italy, then they would revise their expectations of future profitability downwards in response to the lockdown. Therefore, we would see that treated firms as a whole perform worse than control firms.

If investors believe that early discussions of the pandemic implied that these firms were better prepared to weather the storm, then their returns would be better than those that seemed to have been caught "off-guard". We hypothesise that the former is likelier than the latter, since it is not clear how firms could have unilaterally prepared for the over-arching extent of the shock (such as to demand disruptions) just a couple of months in advance.

Our **second hypothesis** is that low-profitability firms with higher share of foreign exchange earnings, higher share of inventories, greater dependence on trade credit and higher operating expenses should have witnessed lower stock market returns when the lockdown was announced, compared to more domestically oriented firms which were more profitable, were holding lower inventories, had lower dependence on trade credit and also lower operating expenses.

3.1 Estimation strategy

We use a difference-in-difference strategy to estimate the impact of the lockdown event on firms' stock market returns. We consider all "treated" firms as one group by using a dummy ("Covid dummy"). Our dependent variable is the cumulative abnormal stock market returns (CARs) for each firm over a window of (-1, +2) days around the lockdown event, i.e. between 23 March (Monday) and 26 March (Thursday). To obtain these abnormal returns, we estimate a market model (i.e. controlling for movements in the Nifty50 index), as shown in equation 1, over a period of 81 days prior to March 24. More specifically our window starts 91 days prior to the lockdown and stops 11 days before the lockdown. The advantage of using a tight window around the event is that it better accounts for anticipation effects and other confounding factors.

$$R_{f,t} = \alpha + \beta R_t + \epsilon \quad (1)$$

where R is the daily returns for firm f and on day t . R_t is the daily return on the Nifty 50. We then use the ϵ to obtain cumulative abnormal returns for each firm f over the event window which spans from the day before the event i.e. 23 March 2020 to two days after the event, i.e. 26 March 2020. Once we obtain the firm-specific CARs at the end of the event window, we run the following cross-sectional regression:

$$CAR_f = \alpha + \gamma_{sector} + \beta_1 COVID\ dummy + \beta_1 \log(age)_f + \beta_2 \log(size)_f + \beta_3 X_f + \epsilon_f \quad (2)$$

where γ_{sector} are NSE-sector dummies, and X_f is a set of balance sheet variables. Among the regressors, of particular interest is the *Covid dummy* which takes a value 1 for the "treated" groups of firms. It should be interpreted as the difference in CARs between the "treated" and the "control" firms: a negative value would imply that the treated group has a *lower* CAR around the lockdown event, while a positive coefficient would imply

that the treated group has a *higher* CAR around lockdown. Other regressors include the balance sheet variables shown in table 2 that are introduced one by one above as well as dummy variables for the sectors that the firms belong to.

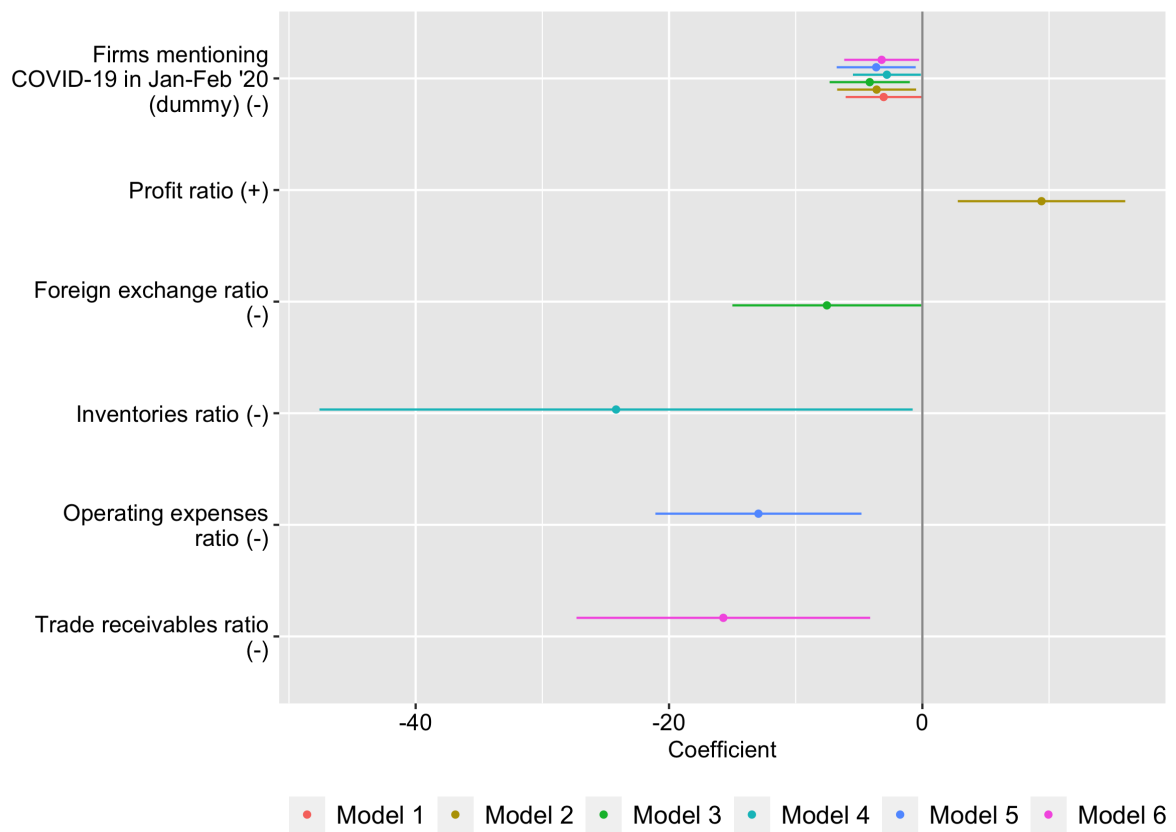
The balance sheet data is as of 31 March, 2019. We use this data for two reasons. Firstly because it is pre-determined and unaffected by the pandemic, thereby reflecting “normal” conditions. Secondly, balance sheet variables are likely to be persistent and would reflect the conditions that these firms *enter* the pandemic with – making them the relevant metrics to consider when evaluating why certain firms may have under or over performed during specific events of the pandemic.

3.2 Results

We summarise our results in figure 3, which are based on the regression results presented in appendix tables 3 and 4. We plot the results from our baseline model (model 1) which includes only age and size of firms, and the sector dummies, over and above the Covid dummy. We also plot the main results that are significant at 90% confidence interval, from models where in addition to the Covid dummy, age, size, and sectors, we sequentially add the regressors of interest: profit, FX earnings, inventories, operating expenses, and trade receivables. We list our main findings below.

1. In all our specifications, the stock returns of "treated" firms, i.e. those that mentioned Covid-19 in their call reports early on in 2020, significantly underperform (at the 90% confidence level or more) the "control" firms. On average, returns of the "treated" firms are roughly 3.5 percentage points lower.
2. We find that the equity returns of more profitable firms outperformed those of less profitable ones by 9 percentage points (model 3). Higher profitability implies higher ability to withstand large revenue shortfalls.
3. Firms with a higher share of foreign exchange earnings in their total income performed worse. They were likely to be more affected due to supply and demand disruptions in the rest of the world.
4. Firms with high inventories saw 24 percentage points lower returns. High share of inventories in total assets might make it difficult for firms to get rid of their inventories once an economy wide lockdown is announced yet they would have had to incur the costs of maintaining these inventories which makes them worse off than firms with lower share of inventories ([Banerjee et al., 2020](#)).
5. Firms with higher pre-pandemic trade credit reliance saw significantly lower abnormal returns. This is likely because in a broad-based crisis such as this one, credit markets are likely to freeze along both extensive and intensive margins. Thus, rolling over existing trade credit as well as obtaining new supply of trade credit would be difficult ([Banerjee et al., 2020](#)).
6. Firms with higher pre-pandemic operating expenses also fared worse once the lockdown was announced. Operating expenses are typically short term expenses. In absence of steady revenues in a lockdown, firms would depend on credit from the financial system to meet these expenses. During a crisis if the financial system is

Figure 3 Explaining cumulative abnormal returns around 24 March lockdown



Note: The dots represent point estimates, and the lines around them are the 90% confidence intervals. Y-axis labels have coefficient signs in brackets. All models include size, age, and sector dummies.

Note: All panels use NSE sector classification. Panel (a) shows the distribution of Covid19 words. For the other three panels, we first extract sentences that are related to Covid19, using the same keywords as those used in panel (a). Then, we construct frequencies of demand, supply, and uncertainty related words within those sentences. Panel (b) shows the distribution of supply-related words (such as “supply”, “supplies”, “supply-chains”, etc) in Covid19 related sentences. Similarly, panels (c) and (d) show distribution of demand and uncertainty related terms in Covid19 sentences.

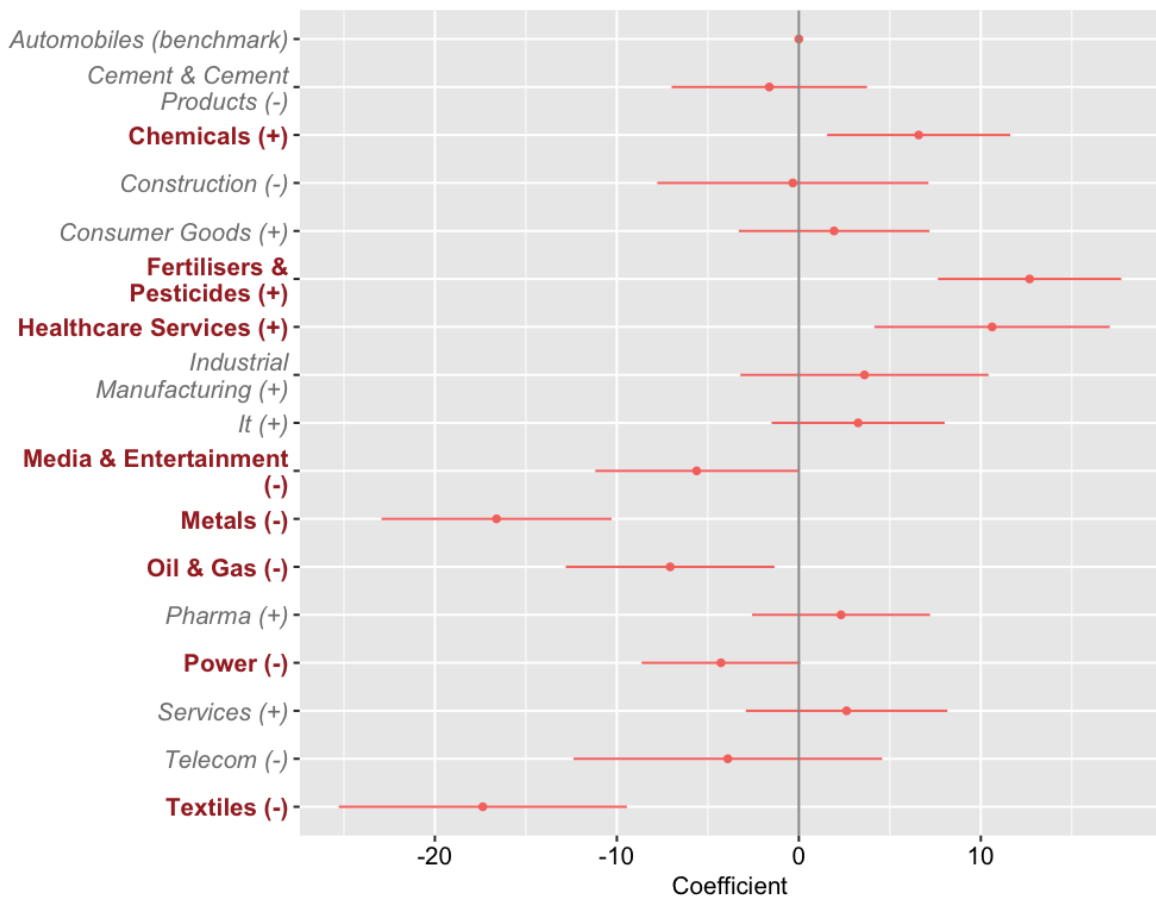
unwilling to offer short term credit (Sengupta and Vardhan, 2020), then these firms are likely to witness lower stock returns.

In figure 4, we plot the coefficients on the sector dummies from the baseline model 1, with only age and size included as controls. Automobiles is the benchmark sector. We find that stock returns of more consumer facing sectors (textiles, media and entertainment) and those that rely on supply chains (metals, and oil and gas) did particularly badly when the lockdown was announced. On the other hand, healthcare services in particular outperformed as compared to automobiles.

In a nutshell, we find that when the nationwide lockdown was announced on 24 March, firms who mentioned Covid-19 in their earnings calls in early-2020 and hence were more exposed to the pandemic, fared worse than firms who did not discuss the pandemic. This result holds when we account for the sectors and key balance sheet characteristics of the firms.

As discussed in Fahlenbrach *et al.* (2020), less financially flexible firms are less able to

Figure 4 Sector dummies from baseline regression



Note: The dots represent point estimates, and the lines around them are the 90% confidence intervals. Red y-axis labels indicate significance at 10% confidence level or more, with coefficient signs in brackets.

Note: All panels use NSE sector classification. Panel (a) shows the distribution of Covid19 words. For the other three panels, we first extract sentences that are related to Covid19, using the same keywords as those used in panel (a). Then, we construct frequencies of demand, supply, and uncertainty related words within those sentences. Panel (b) shows the distribution of supply-related words (such as “supply”, “supplies”, “supply-chains”, etc) in Covid19 related sentences. Similarly, panels (c) and (d) show distribution of demand and uncertainty related terms in Covid19 sentences.

withstand large negative shocks to their revenues, which translates to worse equity market performance. Lower cash, lower profitability, lower diversification in earnings (e.g. higher reliance on foreign exchange revenues) or in borrowing sources (e.g. higher reliance on trade credit) can all be considered indicators of low financial flexibility. In other words, firms that had lower financial flexibility in the pre-pandemic period were worse affected when the lockdown was announced.

We further find that controlling for mentions of “supply” and “demand” related words (not shown here) in the firms’ call reports – which may account for the nature of their exposure to the pandemic – does not change the results qualitatively, and makes them stronger in some specifications.

Firms with more cash holdings reported higher returns on average around the lockdown announcement, but this effect is not significant on average as shown in column (3) of table 3. In further tests in table A.1, particularly column (1), we find some evidence

of non-linearities: firms with above-median cash holdings significantly outperform their counterpart.

4 Analysis of larger sample of listed firms

In ongoing work, we are expanding our event study analysis to the universe of listed non-financial, non-government firms in India. Our sample is approximately 1700 firms listed on the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE). We run the same event study as described in section 3, except now we do not divide the firms into “treatment” and “control” groups. As before, we investigate the firm-specific factors that may have differentially affected the stock market performance of firms around the first lockdown announcement of March 24.

In addition we are also interested in understanding the potential impact on the firms of an unexpected announcement made by the Reserve Bank of India (RBI) just 3 days after the lockdown. On March 27, the RBI made an unscheduled monetary policy announcement wherein the short-term policy interest rate was cut by 75 basis points from to 4.40 per cent from 5.15 per cent with immediate effect, to revive growth and mitigate the impact of Covid-19 on the economy.⁵ The RBI also announced a loan moratorium to provide relief to all categories of borrowers:

*All commercial banks (including regional rural banks, small finance banks and local area banks), co-operative banks, all-India Financial Institutions, and NBFCs (including housing finance companies and micro-finance institutions) (“lending institutions”) are being permitted to allow a moratorium of three months on payment of instalments in respect of all term loans outstanding as on March 1, 2020. Accordingly, the repayment schedule and all subsequent due dates, as also the tenor for such loans, may be shifted across the board by three months.*⁶

Thus, while the March 24 lockdown announcement would have adversely affected nearly all firms in all sectors of the economy, the March 27 announcement of the RBI would have positively impacted at least some types of firms and we try to uncover these differences. In figure 5 we plot the event study results around these two dates.

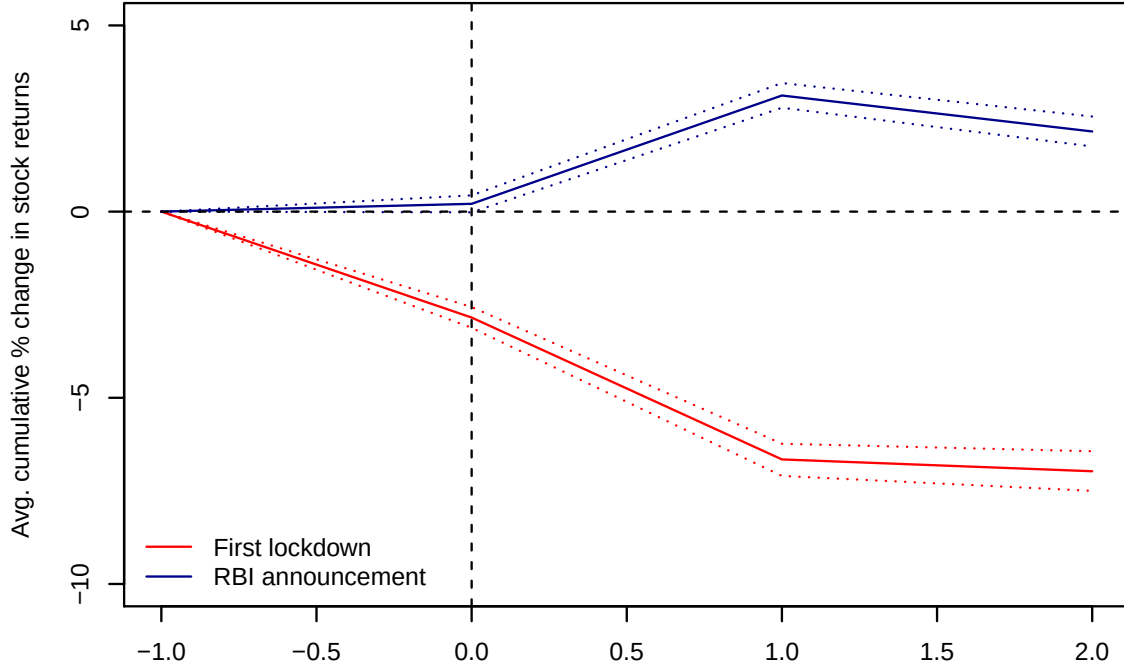
We find that, as expected, while the first event of March 24 resulted in negative abnormal returns of the firms in our sample, the second event of March 27 had an opposite effect producing positive abnormal returns between the day of the event and the next day. This is further demonstrated in tables 5 and 6 which show the average cumulative abnormal returns at the sector levels. While the lockdown announcement led to negative CARs across all sectors, the RBI announcement resulted in positive CARs across the same sectors.

We next conduct a regression analysis wherein we find out the firm-specific heterogeneities in the responses to the two events by regressing the cumulative abnormal returns obtained from the event study on a bunch of pre-pandemic balance sheet variables. The results for

⁵In figure A.1 we plot the change in the daily overnight index swap (OIS) rates around the events to highlight the extent of surprise associated with the RBI’s unscheduled announcement of March 27.

⁶See: https://www.rbi.org.in/Scripts/BS_PressReleaseDisplay.aspx?prid=49582.

Figure 5 Event study around first lockdown (24/03/2020) and RBI meeting (27/03/2020): All firms



Note: The event study follows the same methodology as shown in equation 2, using two events, the first lockdown on 24 March 2020 and the subsequent RBI announcement on 27 March 2020. For both event studies, we use an estimation window of (-91,-11) days before each event and an event window of (-1,+2) around the event date. Each event study is estimated for 1770 firms, who are listed on either NSE and BSE.

the first event are shown in tables 7 to 9.⁷ The dependent variable in these tables is the cumulative abnormal returns of each firm at the end of the (-1,+2) event window around the lockdown announcement on 24 March.

The CARs are calculated using the market model specified in the equation 1, and the cross-sectional regression analysis is done by estimating equation 2. All model specifications in these tables control for the Prowess industry dummies, firm age, firm size (based on last reported annual results), as well as firm market-to-book ratio and relative trading volume (based on data on 20 March 2020).

We find some similarities in our results with the first part of our analysis where we looked at a smaller sample of firms that were presumably more exposed to the pandemic early on. For instance, firms with a higher share of foreign exchange earnings in their total income, firms with higher pre-pandemic operating expenses, and firms with higher pre-pandemic trade credit reliance, fared worse once the lockdown was announced. In addition we find that firms with a higher share of cash balances in their total assets and firms with more collateral were positively affected by the lockdown announcement. This make intuitive sense because cash gives financial flexibility which plays a crucial role in helping firms withstand large shocks to their revenues. Likewise, a firms with more assets that can be pledged as collateral would be better off due to improved access to secured credit.

⁷The estimation for the second event is currently work in progress.

Some of these results stand out even more strongly when we consider the combined effect of a bunch of variables in table 9. We find that the cash variable is consistently important across all model specifications highlighting the vital role of liquidity during such a crisis. In results not reported here, we find that firms at the top 10% of the cash distribution saw significantly higher returns than firms at the bottom 10% of the cash distribution – while those in the middle of the distribution do not stand out. The evidence therefore continues to support the existence of significant non-linearities in the effects of cash. FX ratio is also highly significant with a negative sign, implying that firms with external exposure got particularly hit by the lockdown announcement. In addition we find that firms with high leverage also got adversely impacted which too makes sense because higher the leverage, the lower the financial flexibility of the firms to deal with a liquidity or a revenue shock.

Trade receivables and collateral continue to be important as well, though not as consistently as say cash or FX ratio or leverage. Operating expenses is no longer statistically significant in this table. This underscores the need for further investigation of the effect of some of these variables. By and large our findings remain consistent: less financially flexible firms are less able to withstand large negative shocks to their revenues, which translates to worse equity market performance.

We will next analyse the firm-specific factors that may have affected the response of the stock market returns to the RBI announcement of March 27. It is possible that firms with high leverage benefitted more from this event considering RBI announced a loan moratorium or firms with high operating expenses and hence a more urgent need for short-term credit would also be positively affected because the RBI announced a policy-rate cut. These are some of the hypotheses we plan to examine going forward in this paper.

5 Conclusion

Using the informational content of earnings call reports of some of the largest, non-financial firms in India we throw light on the firms and sectors that may have been more exposed to the pandemic as early as January and February 2020 when as per the official statistics, the disease had still not spread in India. We find that these firms were also worse affected by the announcement of a nationwide lockdown in March compared to firms that were presumably less exposed to the pandemic early on.

Our results highlight the kind of firms that are likely to be more affected when a crisis such as the ongoing one hits the economy. Firms with lower profits, higher share of foreign exchange earnings, higher share of inventories, greater dependence on trade credit and higher operating expenses fared worse on the stock market when the lockdown was announced.

Our ongoing research will throw further light on the kind of firms that were more affected as a result of the national lockdown announcement, from amongst the entire universe of listed firms in India. This kind of a disaggregated firm and sector-level analysis can have important implications for policymakers as they gear up to address corporate vulnerabilities resulting from the pandemic and the associated lockdown.

References

- Banerjee R, Illes A, Kharroubi E, Garralda JMS (2020). “Covid-19 and corporate sector liquidity.” *BIS Bulletins* 10, Bank for International Settlements. URL <https://ideas.repec.org/p/bis/bisblt/10.html>.
- Bansal A, B G, Jacob J, , Srivastava P (2020). “When the market went viral: COVID-19, stock returns, and firm characteristics.” Available at SSRN (21 Jun, 2020). URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3626273.
- Borochin PA, Cicon JE, DeLisle J, Price SM (2018). “The effects of conference call tones on market perceptions of value uncertainty.” *Journal of Financial Markets*, **40**(C), 75–91. URL <https://EconPapers.repec.org/RePEc:eee:finmar:v:40:y:2018:i:c:p:75-91>.
- Bushee BJ, Matsumoto DA, Miller GS (2003). “Open versus closed conference calls: the determinants and effects of broadening access to disclosure.” *Journal of Accounting and Economics*, **34**(1-3), 149–180. URL <https://EconPapers.repec.org/RePEc:eee:jaecon:v:34:y:2003:i:1-3:p:149-180>.
- Bushee BJ, Matsumoto DA, Miller GS (2004). “Managerial and investor responses to disclosure regulation: The case of Reg FD and conference calls.” *The Accounting Review*, **79**(3), 617–643. URL <https://aaapubs.org/doi/10.2308/accr.2004.79.3.617?mobileUi=0>.
- Fahlenbrach R, Rageth K, Stulz RM (2020). “How valuable is financial flexibility when revenue stops? Evidence from the COVID-19 crisis.” *NBER Working Papers* 27106, National Bureau of Economic Research. URL <https://ideas.repec.org/p/nbr/nberwo/27106.html>.
- Frankel R, Johnson M, Skinner D (1999). “An empirical examination of conference calls as a voluntary disclosure medium.” *Journal of Accounting Research*, **37**(1), 133–150. URL <https://EconPapers.repec.org/RePEc:bla:joares:v:37:y:1999:i:1:p:133-150>.
- Hale T, Webster S, Petherick A, Phillips T, , Kira B (2020). “Oxford COVID-19 Government Response Tracke.” *Dataset*, Blavatnik School of Government. Data use policy: Creative Commons Attribution CC BY standard.
- Hassan TA, Hollander S, van Lent L, Tahoun A (2020). “Firm-level exposure to epidemic diseases: Covid-19, SARS, and H1N1.” *NBER Working Papers* 26971, National Bureau of Economic Research. URL <https://ideas.repec.org/p/nbr/nberwo/26971.html>.
- Mathur A, Sengupta R (2019). “Analysing monetary policy statements of the Reserve Bank of India.” *IHEID Working Papers* 08-2019, Economics Section, The Graduate Institute of International Studies.
- Ramelli S, Wagner AF (2020). “Feverish stock price reactions to COVID-19.” *Swiss Finance Institute Research Paper Series* 20-12, Swiss Finance Institute. Forthcoming at *Review of Corporate Finance Studies*, URL <https://ideas.repec.org/p/chf/rpseri/rp2012.html>.
- Sane R, Sharma A (2020). “Holding their breath: Indian firms in an interruption of revenue.” *The Leap Blog* (03 Apr, 2020). URL <https://blog.theleapjournal.org/2020/04/holding-their-breath-indian-firms-in.html>.
- Sengupta R, Vardhan H (2020). “Policymaking at a time of high risk-aversion.” *Ideas for India* (06 Apr, 2020). URL <https://www.ideasforindia.in/topics/money-finance/policymaking-at-a-time-of-high-risk-aversion.html>.

Tables and figures

Table 3 Baseline results for CARs around first lockdown, sample restricted to only non-financial sectors as per NSE classification (1/2)

	<i>Dependent variable:</i>						
	Cumulative abnormal returns around first lockdown announcement (24/03/2020)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Firm mentions COVID dummy	-1.171 (1.643)	-3.054* (1.778)	-3.020* (1.751)	-3.611* (1.869)	-4.157** (1.861)	-3.777* (2.081)	-3.194* (1.898)
<i>log age</i>		0.980 (1.133)	0.959 (1.124)	1.185 (1.132)	1.519 (1.274)	1.248 (1.178)	0.945 (1.157)
<i>log size</i>		0.372 (0.572)	0.376 (0.582)	0.166 (0.586)	0.104 (0.634)	-0.223 (0.711)	0.459 (0.578)
Cash/TA			2.326 (9.172)				
PBDITA/TA				9.405** (3.917)			
FX earnings/Total income					-7.538* (4.334)		
Borrowings/TA						-4.034 (7.866)	
Total equity/TA							2.635 (4.459)
Automobile				Benchmark sector			
Cement & cement products		-1.624 (3.311)	-1.504 (3.388)	-1.208 (3.497)	2.127 (2.985)	-0.175 (3.981)	-1.478 (3.163)
Chemicals		6.585** (3.001)	6.597** (3.123)	6.184** (3.114)	7.540** (3.693)	9.184** (4.010)	6.819** (3.176)
Construction		-0.329 (4.630)	-0.299 (4.610)	-1.820 (4.518)	-0.350 (4.764)	0.513 (4.568)	0.172 (4.484)
Consumer goods		1.937 (3.067)	1.951 (3.191)	1.683 (3.117)	0.796 (3.309)	0.577 (3.581)	1.993 (3.282)
Fertilisers & pesticides		12.680*** (3.115)	12.788*** (3.156)	12.583*** (3.358)	14.839*** (3.564)	13.194*** (3.706)	13.103*** (3.358)
Healthcare services		10.620*** (3.881)	10.221*** (3.835)	7.825 (4.887)	9.556** (4.338)	6.369** (2.657)	10.082** (3.911)
Industrial manufacturing		3.607 (4.072)	3.643 (4.098)	3.225 (4.115)	1.692 (4.578)	3.642 (4.261)	3.700 (4.118)
IT		3.259 (2.851)	3.064 (2.879)	1.978 (2.921)	9.071** (3.859)	5.126 (3.252)	2.790 (3.125)
Media & entertainment		-5.614 (3.422)	-5.544 (3.472)	-7.831** (3.149)	-7.973** (3.973)	-7.508* (3.995)	-5.720 (3.466)
Metals		-16.611*** (3.741)	-16.507*** (3.902)	-17.246*** (3.605)	-16.305*** (4.057)	-14.828*** (4.069)	-16.407*** (3.971)
Oil & gas		-7.073** (3.368)	-7.088** (3.357)	-7.092** (3.232)	-7.386* (3.748)	-5.288 (3.828)	-6.952** (3.445)
Pharma		2.318 (2.980)	2.313 (3.001)	1.319 (2.956)	5.440 (3.823)	3.120 (3.217)	2.208 (3.024)
Power		-4.279* (2.543)	-4.158 (2.691)	-9.141*** (3.215)	-6.291* (3.457)	-2.315 (3.693)	-3.957 (2.699)
Services		2.625 (3.291)	2.590 (3.302)	2.235 (3.210)	0.881 (3.929)	3.662 (3.808)	2.970 (3.319)
Telecom		-3.904 (5.299)	-3.790 (5.239)	-6.532 (5.415)	-1.255 (6.575)	-2.719 (5.862)	-3.856 (5.258)
Textiles		-17.368*** (5.031)	-17.251*** (4.891)	-17.730*** (4.726)	-17.834*** (4.483)	-16.430*** (5.579)	-16.935*** (5.050)
Constant	-3.828*** (0.997)	-10.919 (7.283)	-11.031 (7.473)	-10.605 (7.242)	-8.221 (8.567)	-4.756 (8.714)	-13.277 (8.405)
Observations	151	151	151	150	128	129	151
Adjusted R ²	-0.004	0.198	0.193	0.213	0.247	0.208	0.194

*p<0.1; **p<0.05; ***p<0.01

Wild bootstrap standard errors clustered at firm-level in parentheses.

Note: This table represents the main results from equation 2. The dependent variable is always the cumulative abnormal returns of each firm at the end of the (-1,+2) event window around the first lockdown announcement on 24 March 2020. It is obtained from the market model in equation 1. Column (1) represents the unconditional difference between the returns of “control” and “treated” firms, as shown in table 2. From column (2) onwards, we include dummies for NSE sector classification, as well as control for firm age and size. From column (3) onwards, we include balance sheet variables of interest one by one. In results not reported here, we further control for mentions of supply and demand in these models, and find that the results remain consistent.

Table 4 Baseline results for CARs around first lockdown, sample restricted to only non-financial sectors as per NSE classification (2/2)

	<i>Dependent variable:</i>				
	Cumulative abnormal returns around first lockdown announcement (24/03/2020)				
	(1)	(2)	(3)	(4)	(5)
Firm mentions COVID dummy	−2.801*	−4.183*	−3.168	−3.645*	−3.215*
	(1.667)	(2.206)	(2.002)	(1.864)	(1.866)
log age	1.211	1.487	1.029	0.937	1.109
	(1.137)	(1.223)	(1.197)	(1.106)	(1.088)
log size	0.398	−0.286	0.367	0.269	0.334
	(0.607)	(0.683)	(0.579)	(0.580)	(0.586)
Inventories/TA	−24.179*				
	(14.300)				
St borrowing/Borrowing		−2.975			
		(2.306)			
Tangibles/TA			−1.327		
			(3.558)		
Operating expenses/total income				−12.941**	
				(5.105)	
Trade receivables/TA					−15.712**
					(7.015)
Observations	151	129	151	151	151
Adjusted R ²	0.243	0.216	0.193	0.221	0.219
Sector FE	Yes	Yes	Yes	Yes	Yes

*p<0.1; **p<0.05; ***p<0.01

Wild bootstrap standard errors clustered at firm-level in parentheses.

Note: This table represents the main results from equation 2. The dependent variable is always the cumulative abnormal returns of each firm at the end of the (−1,+2) event window around the first lockdown announcement on 24 March 2020. It is obtained from the market model in equation 1. All columns control for NSE sector dummies and firm age and size. In results not reported here, we further control for mentions of supply and demand in these models, and find that the results remain consistent.

Table 5 Average stock market returns by aggregated industry classification, around first lockdown (24/03/2020)

	Industry classification	Average CAR
1	Mining	−12.637
2	Communication services	−11.460
3	Others	−11.173
4	Metal	−10.975
5	Construction & real estate	−9.928
6	Diversified manufacturing	−9.631
7	Transport equipment	−8.438
8	Textiles	−7.662
9	Wholesale & retail trading	−6.887
10	Food and agro	−6.765
11	Construction material	−6.367
12	Miscellaneous services	−6.136
13	Transport services	−5.673
14	Miscellaneous manufacturing	−5.672
15	Chemical	−5.538
16	Machinery	−5.471
17	Hotels & tourism	−5.343
18	IT	−5.038
19	Electricity	−4.646
20	Consumer goods	−2.918

Note: This table represents the average cumulative abnormal returns from the larger event study, where the sample is approximately 1800 firms listed on both NSE and BSE. Cumulative abnormal returns are calculated using the market model, i.e. using the equation 1 in the text, around the first lockdown on 24 March 2020. The Prowess industry group codes have been aggregated up industry-wise to ensure readability.

Table 6 Average stock market returns by aggregated industry classification, around RBI announcement (27/03/2020)

	Industry classification	Average CAR
1	Transport equipment	0.189
2	Communication services	0.578
3	Textiles	0.674
4	Metal	0.793
5	Electricity	0.853
6	Miscellaneous manufacturing	0.979
7	Wholesale & retail trading	1.203
8	Construction material	1.358
9	IT	1.408
10	Construction & real estate	1.641
11	Machinery	1.718
12	Hotels & tourism	2.312
13	Miscellaneous services	2.452
14	Chemical	2.864
15	Transport services	3.343
16	Consumer goods	4.053
17	Diversified manufacturing	5.165
18	Mining	5.954
19	Others	6.122
20	Food and agro	6.701

Note: This table represents the average cumulative abnormal returns from the larger event study, where the sample is approximately 1800 firms listed on both NSE and BSE. Cumulative abnormal returns are calculated using the market model i.e. using the equation 1 in the text, around the RBI announcement on 27 March 2020. The Prowess industry group codes have been aggregated up industry-wise to ensure readability.

Table 7 Universal event study: First lockdown (1/3)

	<i>Dependent variable:</i>			
	CAR $(-1, +2)_i$ around first lockdown (24/03/2020)			
	(1)	(2)	(3)	(4)
Cash/TA _i	10.872*** (3.564)			
PBDITA/TA _i		0.024 (0.069)		
FX earnings/Total income _i			-0.693*** (0.207)	
Leverage _i				0.154 (0.095)
Observations	1,726	1,721	1,210	1,543
Adjusted R ²	0.055	0.047	0.069	0.045
log age	Yes	Yes	Yes	Yes
log size	Yes	Yes	Yes	Yes
Market-to-book	Yes	Yes	Yes	Yes
Relative trading volumes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes

*p<0.1; **p<0.05; ***p<0.01
Wild bootstrap standard errors clustered at firm-level in parentheses.

Note: This table represents the main results from the larger event study, where the sample is approximately 1800 firms listed on both NSE and BSE. The dependent variable is always the cumulative abnormal returns of each firm at the end of the (-1,+2) event window around the first lockdown announcement on 24 March 2020. Cumulative abnormal returns are calculated using the market model as in using the equation 1, and the cross-sectional regression as in equation 2. All columns control for Prowess industry dummies and firm age and size (based on last reported annual results), and firm market-to-book ratio and relative trading volume (based on data on 20 March 2020).

Table 8 Universal event study: First lockdown (2/3)

	<i>Dependent variable:</i>					
	CAR (-1, +2) around first lockdown (24/03/2020)					
	(1)	(2)	(3)	(4)	(5)	(6)
Operating expenses/total income	-1.767* (1.065)					
Trade receivables/TA		-5.474** (2.501)				
Collateral			1.447* (0.817)			
Inventories/TA				-3.084 (2.620)		
St borrowing/Borrowing					-1.325 (0.817)	
Tangibles/TA						-0.524 (1.665)
Observations	1,566	1,543	1,688	1,726	1,702	1,712
Adjusted R ²	0.054	0.046	0.046	0.048	0.051	0.048
log age	Yes	Yes	Yes	Yes	Yes	Yes
log size	Yes	Yes	Yes	Yes	Yes	Yes
Market-to-book	Yes	Yes	Yes	Yes	Yes	Yes
Relative trading volumes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes

*p<0.1; **p<0.05; ***p<0.01

Wild bootstrap standard errors clustered at firm-level in parentheses.

Note: This table represents the main results from the larger event study, where the sample is approximately 1800 firms listed on both NSE and BSE. The dependent variable is always the cumulative abnormal returns of each firm at the end of the (-1,+2) event window around the first lockdown announcement on 24 March 2020. Cumulative abnormal returns are calculated using the market model as in using the equation 1, and the cross-sectional regression as in equation 2. All columns control for Prowess industry dummies and firm age and size (based on last reported annual results), and firm market-to-book ratio and relative trading volume (based on data on 20 March 2020).

Table 9 Universal event study: First lockdown (3/3)

	<i>Dependent variable:</i>			
	CAR (-1, +2) around first lockdown (24/03/2020)			
	(1)	(2)	(3)	(4)
Cash ratio	12.086** (4.942)	11.932** (5.104)	11.457** (4.881)	12.341*** (4.757)
FX ratio	-0.567*** (0.197)	-0.629*** (0.226)	-0.547** (0.229)	-0.539** (0.222)
Leverage	-3.187* (1.656)	-3.086* (1.585)	-3.123** (1.549)	-3.868** (1.566)
Operating expenses ratio		-1.348 (2.564)	-0.348 (2.676)	-0.642 (2.613)
Trade receivables ratio			-6.337* (3.461)	-4.908 (3.434)
Collateral				3.074** (1.450)
Observations	1,087	1,087	1,085	1,083
Adjusted R ²	0.076	0.075	0.078	0.082

*p<0.1; **p<0.05; ***p<0.01

Wild bootstrap standard errors clustered at firm-level in parentheses.

Note: This table represents the main results from the larger event study, where the sample is approximately 1800 firms listed on both NSE and BSE. The dependent variable is always the cumulative abnormal returns of each firm at the end of the (-1,+2) event window around the first lockdown announcement on 24 March 2020. Cumulative abnormal returns are calculated using the market model as in using the equation 1, and the cross-sectional regression as in equation 2. All columns control for Prowess industry dummies and firm age and size (based on last reported annual results), and firm market-to-book ratio and relative trading volume (based on data on 20 March 2020).

Appendices

A Additional tables and figures

Table A.1 Investigating the role of cash

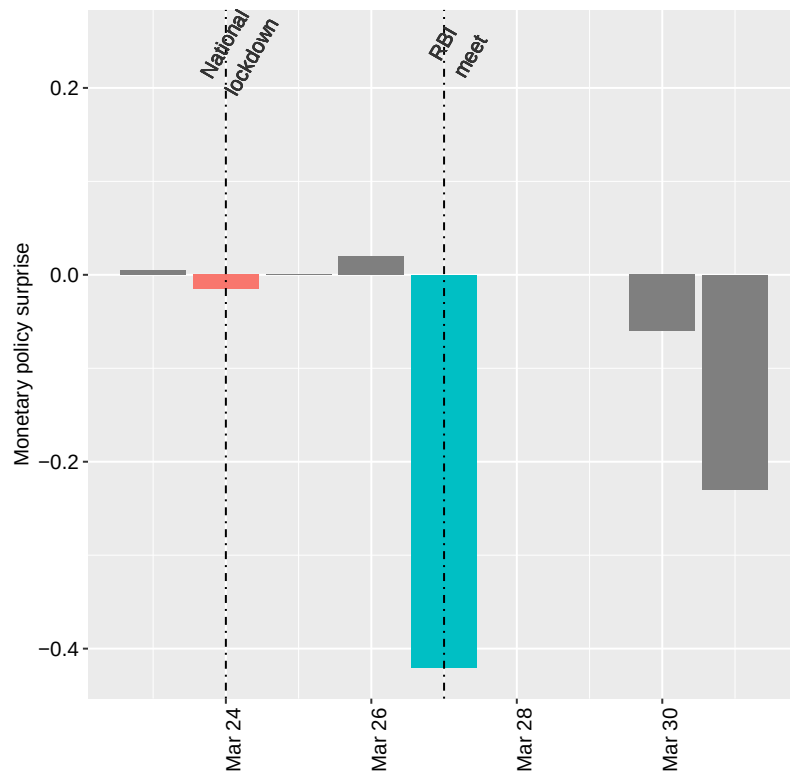
	<i>Dependent variable:</i>				
	Cumulative abnormal returns around first lockdown announcement (24/03/2020)				
	(1)	(2)	(3)	(4)	(5)
Firm mentions COVID dummy	−2.547 (1.713)	−2.820 (1.733)	−2.940* (1.727)	−3.458* (1.930)	−3.070 (1.887)
log age	0.917 (1.121)	0.884 (1.105)	0.956 (1.119)	0.977 (1.094)	0.977 (1.176)
log size	0.449 (0.596)	0.437 (0.584)	0.401 (0.602)	0.425 (0.559)	0.373 (0.591)
Dummy for below median cash	−3.656** (1.580)				
Dummy for below tercile cash		−2.587 (1.681)			
Dummy for below first quantile cash			−2.690 (1.852)		
Dummy for below first percentile cash				−4.499 (4.048)	
Dummy for below 5% cash					−0.169 (3.083)
Observations	151	151	151	151	151
Adjusted R ²	0.223	0.205	0.205	0.208	0.192
Sector FE	Yes	Yes	Yes	Yes	Yes

*p<0.1; **p<0.05; ***p<0.01

Wild bootstrap standard errors clustered at firm-level in parentheses.

Note: In this table, we dig deeper into the role of cash (i.e. liquidity) around the first lockdown announcement. Non-linearities are explored by using dummies for firms with cash that is: below median (column 1), in the bottom 30% (column 2), in the bottom quantile (column 3), below the first percentile (column 4) and below 5% (column 5). There is inadequate variation in cash holdings of firms for all definitions except that in column 1, due to which all coefficients except column 1 are just insignificant at conventional confidence levels. The dependent variable is always the cumulative abnormal returns of each firm at the end of the (-1,+2) event window around the first lockdown announcement on 24 March 2020. It is obtained from the market model in equation 1. All columns control for NSE sector dummies and firm age and size.

Figure A.1 Monetary policy surprises around the national lockdown (24/03/2020) and RBI announcement (27/03/2020)



Note: The chart shows the daily change in the overnight indexed swap (OIS) rate between 23 March 2020 and 31 March 2020. We use this as the proxy for the monetary policy surprise as in [Mathur and Sengupta \(2019\)](#). If the daily change in the OIS rate is close to 0, there is no surprise. The monetary policy surprise on 27 March 2020 is the largest since October 2016.
